

- [2] Steady flow means that the time-rate of change of any quantity in a control volume is zero. +3

→ For these problems we use an instantaneous formulation:

$$\frac{dX_{cv}}{dt} = \dot{X}_{in} - \dot{X}_{out} + \dot{X}_{gen} \quad +3$$

→ 0 for steady-flow assumption +2

→ Examples could vary +2

[3] Given: $v_1 = 10 \text{ m/s}$
 $T_1 = 350 \text{ K}$
 $\rho_1 = 1.225 \text{ kg/m}^3$
 $A_1 = 0.02 \text{ m}^2$
 $c_p = 1001 \text{ J/kg}\cdot\text{K}$
 $v_2 = 20 \rightarrow 200 \text{ m/s}$
 $T_2 = ?$

Assume: ideal gas +1/2
 $c_p = \text{const}$ +1/2
 quasi-steady +1/2

Steady-flow +1/2

$$\Delta p_{re} = \dot{Q} = \dot{W} = 0 \quad +1$$

→ steady flow: +2 $\dot{m} (\Delta h + \Delta ke + \Delta p_{re}) = \dot{Q} - \dot{W}$

$\xrightarrow{+1} \quad \xrightarrow{+1} \quad \xrightarrow{+1}$

$$\Delta h + \Delta ke = 0$$

→ ideal gas / $c_p = \text{const}$: $c_p (T_2 - T_1) + \frac{1}{2} (v_2^2 - v_1^2) = 0$

$\xrightarrow{+1} \quad \xrightarrow{+1}$

$$T_2 = T_1 - \frac{1}{2c_p} (v_2^2 - v_1^2) = 0 \quad +2$$

→ $\dot{m} = \rho_1 v_1 A_1 = (1.225)(10)(0.02)$

$$\boxed{\dot{m} = 0.245 \text{ kg/s}} \quad +1$$

→ Plot → +3 for correct trend

+2 for readability

[4] Given : $P_1 = 5 \text{ MPa}$
 $x_1 = 1$

$P_2 = 3 \text{ MPa}$

Assume : S.C.S. $+1/2$
 : quasi-steady $+1/2$
 : steady flow $+1/2$
 : $\Delta ke = \Delta pe = 0$ $+1/2$
 : $\dot{Q} = 0$ $+1/2$
 : $\dot{W} = 0$ $+1/2$ } throttle

→ steady-flow: $\dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

→ throttle: $\Delta h = 0$ $+2$

$h_2 = h_1$

→ State 1 : $P_1 = 5 \text{ MPa}$

$x_1 = 1$

$h_1 = h_g = 2794.2 \text{ kJ/kg}$ $+2$

→ State 2 : $P_2 = 3 \text{ MPa}$

$h_2 = h_1 = 2794.2 \text{ kJ/kg}$ $+1$

→ $h_f < h_2 < h_g \rightarrow \text{mixture} \rightarrow T_2 = T_{\text{sat}} = 507 \text{ K}$ $+1$

→ $x_2 = \frac{h_2 - h_f}{h_g - h_f} = \frac{2794.2 - 1008.3}{2803.2 - 1008.3} = 0.995$

→ $u_2 = x_2 u_g + (1 - x_2) u_f$ $+1$

$= (0.995)(2603.2) + (0.005)(1004.7)$

$u_2 = 2595.2 \text{ kJ/kg}$ $+1$

[5] Given: Air

H₂O

Process assumptions:

$$T_1 = 623 \text{ K} \quad P_1 = 0.1 \text{ MPa}$$

$$c_p = 1001 \text{ J/kg}\cdot\text{K} \quad x_1 = 0, x_2 = 1$$

$$\dot{m}_{\text{air}} = 0.05 \text{ kg/s} \quad \dot{m}_{\text{H}_2\text{O}} = 0.002 \text{ kg/s}$$

- quasi-g. $+1/2$
- steady flow $+1/2$
- $\Delta ke = \Delta pe = 0$ $+1/2$
- $\dot{W} = 0$ $+1/2$
- no heat loss to outside
- isobaric $+1/2$

Assume: ideal gas $+1/2$ S.C.S. $+1/2$

$$c_p = \text{const} \quad +1/2$$

Water: steady flow $\rightarrow \dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$ $+2$

heat exchanger $\rightarrow \dot{m}(h_2 - h_1) = \dot{Q} + 1$

State 1 $\rightarrow P_1 = 0.1 \text{ MPa}$

$x_1 = 0$

$h_1 = h_f = 417.5 \text{ kJ/kg} \quad +1$

State 2 $\rightarrow P_2 = P_1 = 0.1 \text{ MPa}$

$x_2 = 1$

$h_2 = h_g = 2674.9 \text{ kJ/kg} \quad +1$

$$\Rightarrow \dot{Q} = \dot{m}(h_2 - h_1) = 0.002(2674.9 - 417.5)$$

$$\boxed{\dot{Q} = 4.51 \text{ kW}} \quad +1$$

Air: $\dot{Q}_{\text{air}} = -\dot{Q}_{\text{H}_2\text{O}} = -4.51 \text{ kW} \quad +2$

Steady flow $\rightarrow \dot{m}(\Delta h + \Delta ke + \Delta pe) = \dot{Q} - \dot{W}$

heat exchanger $\rightarrow \dot{m}(h_2 - h_1) = \dot{Q}$

ideal gas / $c_p = \text{const} \rightarrow \dot{m}c_p(T_2 - T_1) = \dot{Q}$

$$T_2 = T_1 + \frac{\dot{Q}}{\dot{m}c_p} = 623 + \frac{-4.51}{(0.05)(1001)}$$

$$\boxed{T_2 = 532.9 \text{ K}} \quad +2$$

6) Given: T_1, P_1, T_2, P_2 } spreadsheet

$$\begin{aligned} \dot{m} &= 30 \text{ kg/s} \\ R &= 287 \text{ J/kg-K} \\ C_v &= 714 \text{ J/kg-K} \end{aligned}$$

Assume: ideal gas $+1/2$
 $C_p, C_v = \text{const}$ $+1/2$
 quasi-steady $+1/2$
 Steady flow $+1/2$
 $\Delta K_e = \Delta P_e = 0$ $+1/2$
 $Q = 0$ $+1/2$

a) steady-flow $\rightarrow \dot{m}(\dot{Q} + \dot{W}) = \dot{Q} - \dot{W}$

compressor $\rightarrow \dot{m}(h_2 - h_1) = -\dot{W}$

Ideal gas / $C_p = \text{const} \rightarrow \dot{m} C_p (T_2 - T_1) = -\dot{W}$ $+2$

$\dot{W} = -\dot{m} C_p (T_2 - T_1)$ where $C_p = C_v + R$

b) plot $\rightarrow$ $+2$ for changing units
 $+2$ for readability
 $+3$ for correct trend

c) Trends $\rightarrow$ $+3$ for any reasonable trends

Compressor $\rightarrow$ compresses fluid ahead of combustor so energy can be extracted in turbine.

$+3$ for any reasonable explanation